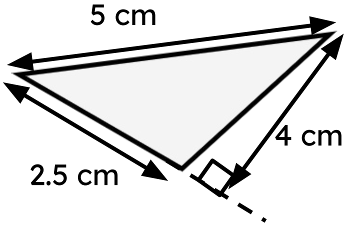
Using the formula for the area of a triangle

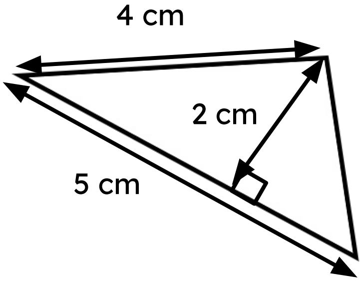
Task A: Calculating the area of a triangle

1) Calculate the area of each triangle.

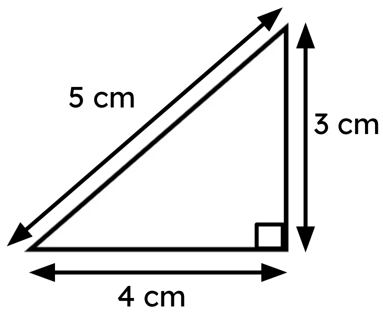
a)



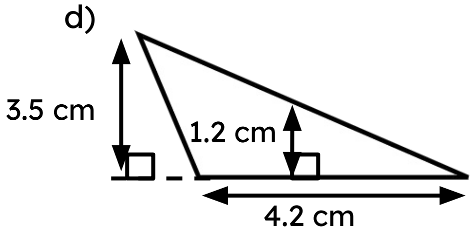
b)



c)



d)



a)
Area = ½ × 2.5 × 4
= 5 cm²

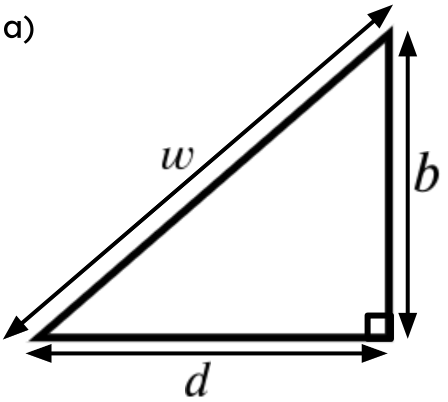
b)
Area = ½ × 2 × 5
= 5 cm²

c)
Area = ½ × 3 × 4
= 6 cm²

d)
Area = ½ × 3.5 × 4.2
= 7.35 cm²

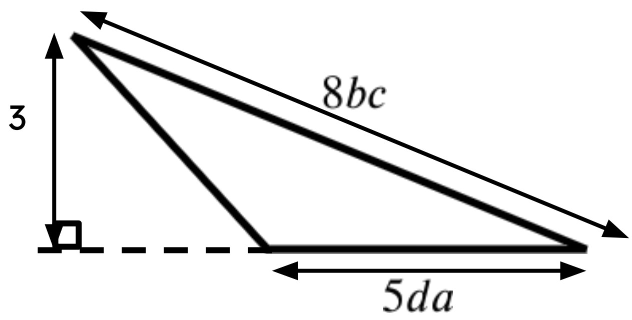
2) Write an expression for the area of each triangle.

a)



Area = $\frac{bd}{2}$

b)

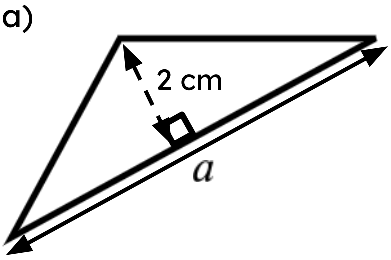


Area = $\frac{15da}{2}$

Task B: Calculating a slide length

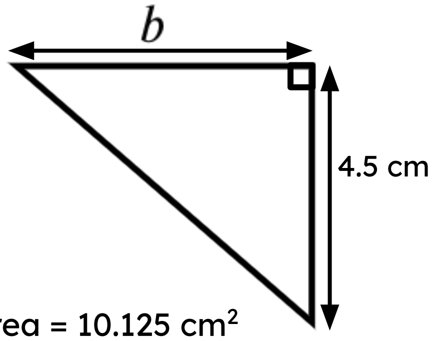
1) Calculate the missing side length for each triangle.

a)



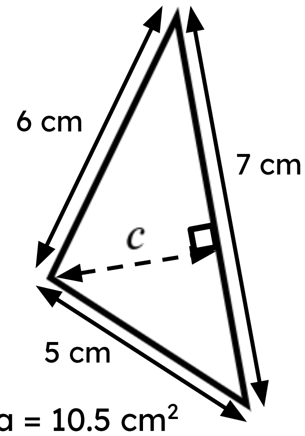
Area = 8 cm^2

b)



Area = 10.125 cm^2

c)



Area = 10.5 cm^2

2) A triangle has an area of 120 cm^2. It has a height of 12 cm. What is the length of its base?

3) Another triangle also has an area of 120 cm^2. What could the lengths of its base and height be? Try to come up with a unique pair of lengths.

a) *The missing length measures 8 cm*

$$8 = \frac{1}{2} \times 2 \times a$$
$$8 = a$$

b) *The missing length measures 4.5 cm*

$$10.125 = \frac{1}{2} \times 4.5 \times b$$
$$10.125 = 2.25 \times a$$
$$4.5 = a$$

c) *The missing length measures 3 cm*

$$10.5 = \frac{1}{2} \times 7 \times c$$
$$10.5 = 3.5 \times c$$
$$3 = c$$

2) *The length of the base is 20 cm*

$$120 = \frac{1}{2} \times 12 \times c$$
$$120 = 6 \times c$$
$$20 = c$$

3) *Any pair of numbers that have a product of 240 will work.*